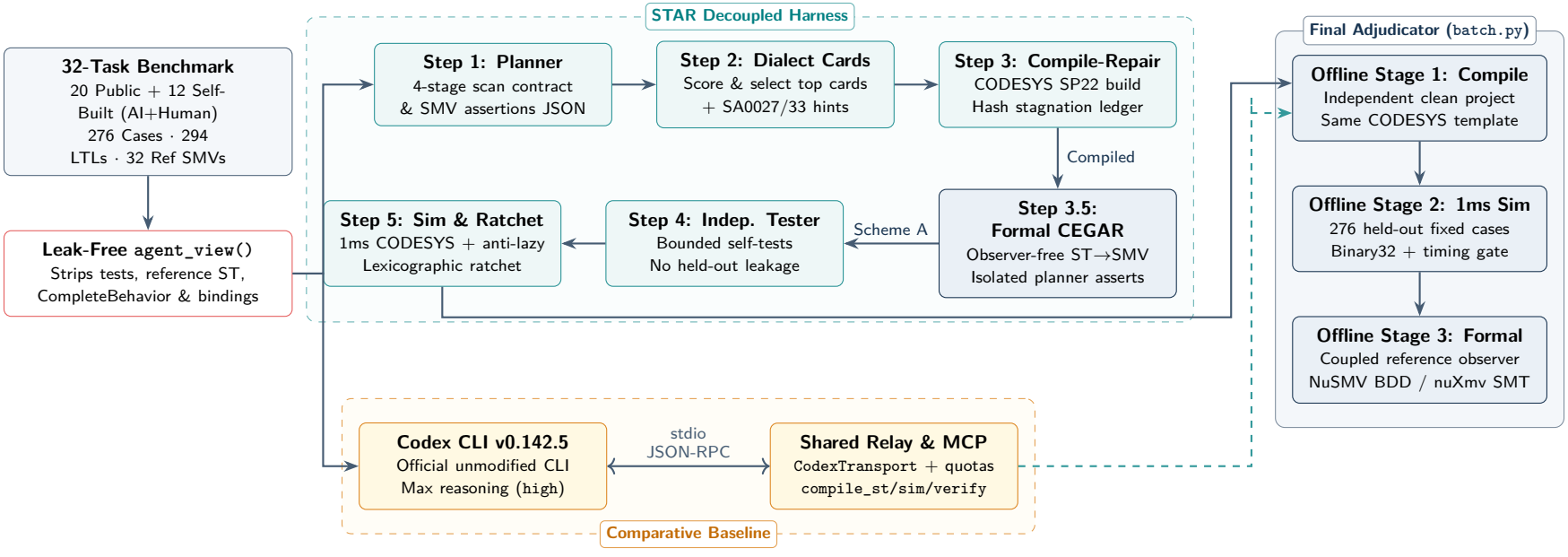

STAR Decoupled Harness
32-Task Benchmark
20 Public + 12 Self-Built (AI+Human)
276 Cases · 294 LTLs · 32 Ref SMVs
Leak-Free agent_view()
Strips tests, reference ST, CompleteBehavior & bindings
Step 1: Planner
4-stage scan contract & SMV assertions JSON
Step 2: Dialect Cards
Score & select top cards + SA0027/33 hints
Step 3: Compile-Repair
CODESYS SP22 build
Hash stagnation ledger
Compiled
Step 3.5: Formal CEGAR
Observer-free ST→SMV
Isolated planner asserts
Scheme A
Step 4: Indep. Tester
Bounded self-tests
No held-out leakage
Step 5: Sim & Ratchet
1ms CODESYS + anti-lazy
Lexicographic ratchet
Final Adjudicator (batch.py)
Offline Stage 1: Compile
Independent clean project
Same CODESYS template
Offline Stage 2: 1ms Sim
276 held-out fixed cases
Binary32 + timing gate
Offline Stage 3: Formal
Coupled reference observer
NuSMV BDD / nuXmv SMT
Codex CLI v0.142.5
Official unmodified CLI
Max reasoning (high)
stdio
JSON-RPC
Shared Relay & MCP
CodexTransport + quotas
compile_st/sim/verify
Comparative Baseline